

Cayley, *Collected Mathematical Papers*, Vol. V

PDF pp. 401–425

Paper 350 (continued): *On the Classification of Cubic Curves*, Arts.
73–122

This installment continues Paper 350 from PDF p. 400, beginning with Article 73 (the equation of the envelope for the Hyperbolas A Defective) and runs through the close of the Memoir at p. 399 of the original printing (Cambridge, February 8, 1864), together with the blank page and the figure-plate “Plate III”.

Paper 350 (continued): On the Classification of Cubic Curves

We may fix the absolute magnitudes of the coordinates by writing $x+z=1$; and the equation then becomes

$$(x^2 + y^2)(1 - x) + 4k\{(\lambda - \nu)x + \mu y + \nu\} = 0.$$

The origin is at the asymptote-point, or intersection of the imaginary asymptotes; the equation of the real asymptote is $z=1$; that of the imaginary asymptotes is $x^2 + y^2 = 0$.

If x and y are ordinary rectangular coordinates then the pair of lines represented by this equation will be an indefinitely small circle, and conversely, if the Asymptote-Point be an indefinitely small circle, then x and y will be rectangular coordinates; and we may without loss of generality assume that this is so.

73. The equation of the envelope is

$$\sqrt{\frac{1}{2}(x + yi)} + \sqrt{\frac{1}{2}(x - yi)} + \sqrt{z} = 0;$$

this gives successively

$$\sqrt{\frac{1}{2}(x + yi)} + \sqrt{\frac{1}{2}(x - yi)} = -\sqrt{z},$$

$$\frac{1}{2}(x + yi) + \frac{1}{2}(x - yi) + 2\sqrt{\frac{1}{2}(x + yi) \cdot \frac{1}{2}(x - yi)} = z,$$

that is

$$x + 2\sqrt{\frac{1}{4}(x^2 + y^2)} = z, \quad \text{i.e.} \quad x + \sqrt{x^2 + y^2} = z;$$

and proceeding further,

$$(x + z)^2 + x^2 + y^2 - 2(x + z)\sqrt{x^2 + y^2} = 4z(x + \sqrt{x^2 + y^2}),$$

$$(x - z)^2 + x^2 + y^2 = 2(x + 3z)\sqrt{x^2 + y^2},$$

that is

$$\{(x - z)^2 + x^2 + y^2\}^2 = 4(x + 3z)^2(x^2 + y^2);$$

$$(x - z)^4 + 2(x^2 + y^2)\{(x - z)^2 - 2(x + 3z)^2\} + (x^2 + y^2)^2 = 0.$$

Putting $z = 1 - x$, and therefore $x - z = 2x - 1$, and $x + 3z = 3 - 2x$, this becomes

$$(2x - 1)^4 + 2(x^2 + y^2)\{(2x - 1)^2 - 2(3 - 2x)^2\} + (x^2 + y^2)^2 = 0,$$

that is

$$(x^2 + y^2)^2 - 2(x^2 + y^2)(4x^2 - 20x + 17) + (2x - 1)^4 = 0,$$

which may also be written

$$y^4 - 2y^2(3x^2 - 20x + 17) + 9x^4 + 8x^3 - 10x^2 - 8x + 1 = 0,$$

or, what is the same thing,

$$y^4 - 2y^2(x-1)(3x-17) + (x-1)(9x-1)(x+1)^2 = 0,$$

for the equation of the envelope. The solution of the equation in y gives

$$y^2 = (x-1)(3x-17) \pm (2x-3)\sqrt{-32(x-1)}.$$

74. The original curve $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ had the three nodes

$$(2, -\frac{1}{2}, -\frac{1}{2}), \quad (-\frac{1}{2}, 2, -\frac{1}{2}), \quad (-\frac{1}{2}, -\frac{1}{2}, 2);$$

and thence writing $\frac{1}{2}(x+yi)$, $\frac{1}{2}(x-yi)$, z for x , y , z , we find the nodes

$$(x = \frac{3}{2}, y = \frac{5}{2}i), \quad (x = \frac{3}{2}, y = -\frac{5}{2}i), \quad (x = -1, y = 0);$$

the first and second of these are acnodes, or the curve has a pair of imaginary acnodes; the third is a crunode; and to find the directions at this point, if in the equation of the curve we write $x-1$ for x , the equation becomes

$$y^4 - 2y^2(x-2)(3x-20) + (x-2)(9x-10)x^2 = 0;$$

the lowest terms therefore are $20(-4y^2 + x^2) = 0$; and we have $y = \pm \frac{1}{2}(x+1)$ for the equation of the tangents at the crunode.

$y = 0$ gives $x = 1$, $x = \frac{1}{9}$ and (as a twofold value) $x = -1$, which belongs to the crunode.

$x = 1$ gives $y^4 = 0$, or the line $x = 1$ is a tangent of four-pointic intersection.

$x = 0$ gives $y^4 - 34y^2 + 1 = 0$, that is $y^2 = 17 \pm 12\sqrt{2}$, or $y = \pm(3 \pm 2\sqrt{2})$.

75. The curve has a pair of asymptotic parabolas, and taking for the equation of one of them

$$(y - x\sqrt{3} + \beta)^2 = -32(x + \alpha),$$

this gives

$$y = x\sqrt{3} - \beta + \sqrt{-32(x + \alpha)},$$

and thence

$$y^2 = 3x^2 - 2\beta x\sqrt{3} + \beta^2 - \frac{2\beta}{\sqrt{3}}(\text{coefficient terms}) + 2\left(x - \frac{\beta}{\sqrt{3}}\right)\sqrt{-32(x + \alpha)},$$

that is, more explicitly,

$$y^2 = 3x^2 - 2\beta x\sqrt{3} + \beta^2 + 2\left(x - \frac{\beta}{\sqrt{3}}\right)\sqrt{-32(x + \alpha)},$$

which agrees with the value of y^2 in the envelope as to the terms x^2 , $x\sqrt{x}$, and by properly determining α , β , it may be made to agree as to the terms x and x^0 .

We have in the parabola

$$y^2 = 3x^2 - 2\beta x\sqrt{3} + \beta^2 + \sqrt{-32}\left\{2x\sqrt{x} + \left(\alpha - \frac{\beta}{\sqrt{3}}\right)\sqrt{x}\right\},$$

and in the curve

$$y^2 = 3x^2 - 20x + 17 + \sqrt{-32}\{2x\sqrt{x} - 4\sqrt{x}\}.$$

We have therefore

$$-2\beta\sqrt{3} = -20, \quad \alpha - \frac{\beta}{\sqrt{3}} = -4,$$

giving

$$\beta = \frac{10}{3\sqrt{3}} = \frac{10\sqrt{3}}{9}, \quad \alpha = -\frac{2}{9},$$

so that the equation of the parabola is

$$(y - (x - \frac{1}{3})\sqrt{3})^2 = -\frac{32}{9}(x - \frac{2}{9});$$

that of the other parabola is of course obtained by merely changing the sign of $\sqrt{3}$.

76. From the foregoing results we may trace the curve, but this may be done somewhat more easily by means of polar coordinates, viz. writing $x = r \cos \theta$, $y = r \sin \theta$, and therefore $z = 1 - r \cos \theta$, we have

$$\sqrt{\frac{1}{2}r} \cdot 2 \cos \frac{1}{2}\theta + \sqrt{1 - r \cos \theta} = 0,$$

that is

$$r(\cos \theta + 8 \cos^2 \frac{1}{2}\theta) = 1;$$

or since

$$\cos \theta = 1 - 8 \sin^2 \frac{1}{2}\theta \cos^2 \frac{1}{2}\theta = 1 - 8 \cos^2 \frac{1}{2}\theta + 8 \cos^4 \frac{1}{2}\theta,$$

we have

$$\cos \theta + 8 \cos^2 \frac{1}{2}\theta = (1 - 4 \cos^2 \frac{1}{2}\theta)^2 = (-1 - 2 \cos \frac{1}{2}\theta)^2,$$

and the equation is

$$r = \frac{1}{(1 + 2 \cos \frac{1}{2}\theta)^2}.$$

$\theta = 0^\circ$ gives $r = \frac{1}{9}$, $\theta = 180^\circ$ gives $r = 1$, $\theta = 240^\circ$ gives $r = \infty$, $\theta = 360^\circ$ gives $r = 1$, values which agree with the results obtained by rectangular coordinates.

77. The form is shown in the figure; we see that the curve consists of a lower branch without any singularity; and of an upper branch which cuts itself in the crunode.

78. The equation of the twofold centre locus (making in the form $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$, the foregoing transformation) is

$$\sqrt{\frac{1}{2}(x + yi)} + \sqrt{\frac{1}{2}(x - yi)} + \sqrt{1 - x} = 0,$$

that is

$$x + \sqrt{x^2 + y^2} = 1 - x,$$

or

$$\sqrt{x^2 + y^2} = 1 - 2x,$$

which is

$$3x^2 - 4x - y^2 = -1,$$

or finally

$$9(x - \frac{2}{3})^2 - 3y^2 = 1,$$

which is a hyperbola having its centre at the harmonic point $x = \frac{2}{3}$, $y = 0$; having $x = \frac{1}{3}$, $x = 1$ for the extremities of the transverse axis, and such that the asymptotes are inclined to the axis of x at an angle $= 60^\circ$; this curve is also shown in the figure.

79. Similarly making the transformation in the equation of the one-with-twofold centre locus written under the form $-(-x + y + z)(x - y + z)(x + y - z) + xyz = 0$, this becomes

$$-(-\frac{1}{2}(x + yi) + \frac{1}{2}(x - yi) + z)(\frac{1}{2}(x + yi) - \frac{1}{2}(x - yi) + z)(\frac{1}{2}(x + yi) + \frac{1}{2}(x - yi) - z) + \frac{1}{4}(x + yi)(x - yi)z = 0,$$

that is

$$-(-yi + z)(yi + z)(x - z) + \frac{1}{4}(x + yi)(x - yi)z = 0,$$

which is

$$-\frac{1}{4}(y^2 + z^2)(x - z) + \frac{1}{4}(x^2 + y^2)z = 0,$$

$$-4z^2(x - z) + x^2z + y^2(5z - 4x) = 0,$$

or, what is the same thing,

$$z(x - 2z)^2 + y^2(5z - 4x) = 0,$$

or, putting for z its value $= 1 - x$, this is

$$(1 - x)(3x - 2)^2 + y^2(5 - 9x) = 0,$$

that is

$$y^2 = -\frac{(x - 1)(3x - 2)^2}{9x - 5},$$

which is the equation of the one-with-twofold centre locus.

80. The curve is symmetrical in regard to the axis of x . And moreover

$$\begin{aligned} x < \frac{5}{9}, & \quad y \text{ is impossible,} \\ x = \frac{5}{9}, & \quad y^2 = \infty, \text{ or the line } x = \frac{5}{9} \text{ is an asymptote,} \\ x = \frac{2}{3}, & \quad y^2 = 0, \text{ which is a crunode,} \\ x = 1, & \quad y^2 = 0, \\ x > 1, & \quad y \text{ is impossible.} \end{aligned}$$

The equation of the tangents at the crunode are $y^2 = 3(x - \frac{2}{3})$, or the tangents are inclined to the axis of x at angles $= 60^\circ$. The curve consists, as shown in the figure, of a single branch cutting itself in the crunode, and tending on each side towards the asymptote.

81. The equation of the harmonic conic, making the foregoing transformation in the equation $\frac{\lambda - \nu}{x} + \frac{\mu}{y} + \frac{\lambda - \mu}{z} = 0$, becomes

$$\frac{\lambda - \nu + \mu i}{x + yi} + \frac{\lambda - \nu - \mu i}{x - yi} + \frac{\mu i - \mu i}{z} = 0,$$

that is

$$-2(\lambda - \nu)yi + 2\mu i x - \mu i \frac{x^2 + y^2}{z} = 0,$$

which is, or what is the same thing,

$$\begin{aligned} \mu(x^2 + y^2) - 2z\{\mu x - (\lambda - \nu)y\} &= 0, \\ \mu(x^2 + y^2 - 2zx) + 2(\lambda - \nu)yz &= 0, \end{aligned}$$

which, putting for z its value $= 1 - x$, is

$$\mu(3x^2 + y^2 - 2x) + 2(\lambda - \nu)y(1 - x) = 0,$$

or developing,

$$3\mu x^2 - 2(\lambda - \nu)xy + \mu y^2 - 2\mu x + 2(\lambda - \nu)y = 0,$$

either of which is the equation of the harmonic conic corresponding to the satellite line $(\lambda - \nu)x + \mu y + \nu = 0$, or, since the direction is alone material, to the satellite line $(\lambda - \nu)x + \mu y = 0$. The second form shows that the conic is

$$\begin{aligned} \text{an ellipse for } & 3\mu^2 > (\lambda - \nu)^2, \\ \text{a parabola „ } & 3\mu^2 = (\lambda - \nu)^2, \\ \text{a hyperbola „ } & 3\mu^2 < (\lambda - \nu)^2. \end{aligned}$$

82. The first form shows that the conic passes through the four points which are the intersection of the ellipse $3x^2 + y^2 - 2x = 0$, (or, as the equation may also be written, $9(x - \frac{1}{3})^2 + 3y^2 = 1$), with the pair of lines $(x - 1)y = 0$: this is right, for the points in question are the three points $(x = 0, y = 0)$, $(x = 1, y = i)$, $(x = 1, y = -i)$, which are the vertices of the triangle formed by the asymptotes $(x^2 + y^2)(x - 1) = 0$; and the point $x = \frac{2}{3}, y = 0$, which is the harmonic point.

83. Putting for shortness $\lambda - \nu = \kappa$, so that the equation of the satellite line is $\kappa x + \mu y + \nu = 0$, and that of the corresponding harmonic conic is

$$\mu(3x^2 + y^2 - 2x) + 2\kappa y(1 - x) = 0,$$

the coordinates of the centre are found from the formulae

$$\frac{1}{2}(x + yi) : \frac{1}{2}(x - yi) : z = (\kappa + \mu i)^2 : (\kappa - \mu i)^2 : -4\mu^2,$$

whence, since $x + z = 1$, we have for the coordinates of the centre

$$x = \frac{\mu^2 - \kappa^2}{3\mu^2 - \kappa^2}, \quad y = \frac{-2\mu\kappa}{3\mu^2 - \kappa^2},$$

and it is easy to verify that these belong to a point on the twofold centre conic

$$3x^2 - 4x - y^2 + 1 = 0.$$

84. The asymptotes of the harmonic conic meet at the centre; and they again cut the twofold centre conic in two points, the intersections of the last-mentioned conic with the line

$$(\kappa + \mu i) \frac{1}{2}(x + yi) + (-\kappa + \mu i) \frac{1}{2}(x - yi) - 2\mu i z = 0,$$

that is

$$\mu x + \kappa y - 2\mu z = 0,$$

or, what is the same thing,

$$3\mu x + \kappa y - 2\mu = 0.$$

85. I remark that the equation of the asymptotes of the harmonic conic is

$$(3\mu^2 - \kappa^2)[\mu(3x^2 + y^2 - 2x) + 2\kappa y(1 - x)] + \mu(\mu^2 + \kappa^2) = 0,$$

and that the theorem for the construction of the asymptotes depends on the identical equation

$$\begin{aligned} (3\mu^2 - \kappa^2)[\mu(3x^2 + y^2 - 2x) + 2\kappa y(1 - x)] + \mu(\mu^2 + \kappa^2) + 3\mu(\mu^2 + \kappa^2)(3x^2 - 4x + 1) \\ = -2[-(3\mu^2 - \kappa^2)x + 2\mu\kappa y + \mu^2 + \kappa^2](3\mu x + \kappa y - 2\mu), \end{aligned}$$

which is easily verified; and where

$$-(3\mu^2 - \kappa^2)x + 2\mu\kappa y + \mu^2 + \kappa^2 = 0$$

is the equation of the tangent of the twofold centre conic $3x^2 - y^2 - 4x + 1 = 0$ at the centre of the harmonic conic.

86. On account of the symmetry in regard to the axis of x , it will be sufficient to attend to the series of curves corresponding to a direction $y = x$ of the satellite line, for which the ratio $-\frac{\kappa}{\mu}$ has a given sign; and the inclination of the satellite line to the asymptote will pass from 90° to 0° according as the value of the ratio $-\frac{\kappa}{\mu}$ passes from 0 to ∞ .

87. First, if $-\frac{\kappa}{\mu}$ is $= 0$, that is, if the satellite line be perpendicular to the asymptote, then the harmonic conic is the ellipse

$$3x^2 + y^2 - 2x = 0,$$

or, as it may also be written,

$$9(x - \frac{1}{3})^2 + 3y^2 = 1.$$

As $-\frac{\kappa}{\mu}$ increases from 0 to $\sqrt{3}$, that is, as the inclination of the satellite line diminishes from 90° to 30° , the harmonic conic becomes an ellipse of greater and greater excentricity, and ultimately a parabola.

88. I notice the particular case $-\frac{\kappa}{\mu} = \frac{1}{2}$, which corresponds to the direction parallel to one of the nodal tangents of the envelope: the harmonic conic is in this case the ellipse

$$3x^2 + xy + y^2 - 2x - y = 0,$$

which, it will be observed, passes through the point $(x = 0, y = 1)$ which is one of the points of intersection of the twofold centre locus or hyperbola $3x^2 - 4x - y^2 = -1$ by the line $x = 0$.

89. For the value $-\frac{\kappa}{\mu} = \sqrt{3}$ corresponding to a direction inclined at an angle $= 30^\circ$ to the asymptote, the harmonic conic becomes the parabola $(x\sqrt{3} - y)^2 - 2(x - y\sqrt{3}) = 0$: this equation may also be written $(x - \frac{1}{2})^2 + y^2 = \frac{1}{4}(x + y\sqrt{3} - 1)^2$, a form which puts in evidence the focus and directrix of the parabola.

90. For a value $-\frac{\kappa}{\mu} > \sqrt{3}$, that is, when the satellite line is inclined to the asymptote at an angle $< 30^\circ$, the harmonic conic is a hyperbola, and ultimately when $-\frac{\kappa}{\mu} = \infty$, or the satellite line is parallel to the asymptote, the harmonic conic becomes the pair of lines $y(1 - x) = 0$.

91. I have in the figure shown the following forms of the harmonic conic; viz.

- hyperbola, corresponding to inclination $< 30^\circ$ of satellite line to asymptote.
- parabola, to inclination $= 30^\circ$.
- ellipse } to inclination = inclination $(= \tan^{-1} 2)$ of a nodal tangent of the envelope.
- ellipse } $>$

and for these forms respectively the successive positions of the satellite line are indicated as follows:

92. For the inclination $< 30^\circ$, the positions are $ACMC'DEA'$, viz.

- A , at infinity,
- C , touching lower branch of envelope,
- M , between C and C' ,
- C' , touching upper branch of envelope,
- D , passing through asymptote point D_1 ,
- E , passing through crunode of envelope,
- A' , at infinity.

The corresponding positions of the critic centres on the hyperbola are

On one branch of hyperbola.	On the other branch.
A_1, A_2 , each at infinity,	A_3 , the harmonic point,
C_{12} , a twofold centre,	C_3 , a one-with-twofold centre,
	M_1 , a real centre,
C'_1 , a one-with-twofold centre and	C'_{23} , a twofold centre,
D_1 (Asymptote-Point), D_2 ;	D_3 ,
E_1 and E_2 ;	E_3 , and harmonic point,
A'_1 , at infinity,	A'_2, A'_3 .

93. For inclination $= 30^\circ$ the positions are $(AC)MC'DE(A'C)$, viz.

- AC , at infinity, touches envelope at infinity,
- M , between (AC) and C' ,
- C' , touching upper branch of envelope,
- D , passing through asymptote point D_1 ,
- E , passing through crunode of envelope,
- $A'C$, at infinity, touches envelope at infinity.

The corresponding positions of the critic centres on the parabola are

- A_3 (harmonic point) a one-with-twofold centre; $A_{12} = A'_{12}$ at infinity, a twofold centre,
- M_1 , real centre, the other two centres being imaginary,

- C'_{23} , a twofold centre; C'_1 , a one-with-twofold centre,
- D_1 (asymptote point), D_2 ; D_3 ,
- $E_1, E_2; E_3$,
- $A'_{12} = A_{12}$, at infinity, a one-with-twofold centre; A'_3 (harmonic point) a twofold centre.

94. For inclination $= \tan^{-1} 2$, the positions are $AMC'D(EC'')NA'$, viz.

- A , at infinity,
- M , between A and C' ,
- C' , touching upper branch of envelope,
- D , through asymptote point D_1 ,
- EC''_1 , touching upper branch of envelope at crunode N between EC'' and A' ,
- A' , at infinity.

And the corresponding positions of the critic centres on the ellipse are

- A_3 (harmonic point), the other two centres imaginary,
- M_1 , real centre, the other two centres imaginary,
- C'_{12} , a twofold centre, C'_3 , a one-with-twofold centre,
- D_1 (asymptote point), D_2 ; D_3 ,
- $C'E_1$, twofold centre; CE_2 , one-with-twofold centre,
- N_1 , real centre, the other two centres imaginary,
- A'_2 , harmonic point; the other two centres imaginary.

95. And for the inclinations $<$ and $> \tan^{-1} 2$, the only difference is that the positions are $ACDEC''A'$, viz.

- A , at infinity,
- M , between A and C' ,
- C' , touching upper branch of envelope,
- D , through asymptote point D_1 ,
- E , through crunode,
- C'' , touching upper branch of envelope,
- N , between C'' and A' ,
- A' , at infinity,

and that instead of the points $C'E_1$ and $C''E_2$ we have separately the points C'_{12} , C''_1 , and E_1, E_2, E_3 , as shown in the figure.

96. For the better understanding of the figure it is to be observed that the points D_2 and D_3 lie on the line $x = \frac{1}{2}$: this depends on the theorem No. 81 of the Memoir on Involution, viz. of the critic centres which belong to a satellite line through the vertex $(0, 0, 1)$, one is the vertex itself, the other two lie on the line $x + y - z = 0$; or making the transformation $\frac{1}{2}(x + iy), \frac{1}{2}(x - iy), z$ for x, y, z and writing $x + z = 1$, of the critic centres which pass through the vertex $(0, 0, 1)$ (the asymptote point), one is this point itself, the other two lie on the line $x - z = 0$, that is on $x = \frac{1}{2}$.

97. Again it is to be observed that the centre E_1 lies on the line $x = 0$, and the centres E_2, E_3 on the circle (indicated by a dotted line in the figure) $(x + \frac{1}{2})^2 + y^2 = \frac{5}{4}$: this depends on the theorem Nos. 73 and 74 of the Memoir on Involution, viz. of the critic centres for satellite lines through the node (acnode) $(1, 1, -4)$, one lies on the line $x + y = 0$, and the other two lie on the conic $z(x + y + z) - xy = 0$; making the substitution

$$\frac{1}{2}(x + yi), \quad \frac{1}{2}(x - yi), \quad z \quad \text{for} \quad x, y, z,$$

and writing also $x + z = 1$, we find that, for the satellite lines which pass through the crunode $(1, 0, 2)$, the critic centres lie one on the line $x = 0$, the other two on the conic

$$z(x + z) - x^2 - y^2 = 0,$$

that is, on the circle $x^2 + y^2 + x - 1 = 0$, or $(x + \frac{1}{2})^2 + y^2 = \frac{5}{4}$.

98. The circle in question cuts the twofold centre conic $3x^2 - 4x - y^2 = -1$ at its intersections with the line $x = 0$, viz. in the points $x = 0, y = \pm 1$; and it moreover touches the one-with-twofold centre locus $y^2 = -\frac{(x-1)(3x-2)^2}{9x-5}$ at the point where this same circle meets the ellipse $3x^2 + xy + y^2 - 2x - y = 0$, which is the harmonic conic corresponding to the inclination $\tan^{-1} 2$. In fact, writing down the three equations,

$$\begin{aligned} x^2 + y^2 + x - 1 &= 0, \\ 3x^2 + xy + y^2 - 2x - y &= 0, \\ y^2 &= -\frac{(x-1)(3x-2)^2}{9x-5}, \end{aligned}$$

the first and third equations give

$$-(x-1)(3x-2)^2 + (9x-5)(x^2 + x - 1) = 0,$$

or, reducing,

$$(5x-3)^2 = 0,$$

that is $x = \frac{3}{5}$, and then from the first or third equation $y^2 = \frac{1}{25}$, or $y = \pm \frac{1}{5}$; hence the circle touches the one-with-twofold centre locus at the points

$$(x = \frac{3}{5}, y = \pm \frac{1}{5}),$$

and by means of the second equation we see that the first of these points, viz. the point $x = \frac{3}{5}, y = -\frac{1}{5}$, is a point of the ellipse or harmonic conic $3x^2 + xy + y^2 - 2x - y = 0$.

99. I consider the analytical theory of the case where the satellite line is parallel to the asymptote; this is in fact similar to the theory *ante* Nos. 67–69; writing

$$\frac{1}{2}(x + yi), \quad \frac{1}{2}(x - yi), \quad z, \quad \lambda - \mu i, \quad \lambda + \mu i, \quad \nu$$

in the place of $x, y, z, \lambda, \mu, \nu$, and putting afterwards $\mu = 0$, that is, in the transformed equation $\lambda x + \mu y + \nu z = 0$ writing $\mu = 0$, we find for the satellite line $\lambda x + \nu(1 - x) = 0$; the equation in θ (the factor $\theta + \lambda = 0$ being disregarded) is

$$\theta^2 - \lambda\theta - 2\lambda\nu = 0,$$

and the corresponding critic centres lie on the line $y = 0$, at the distances given by the equation

$$x = \frac{\lambda(1-z)}{z};$$

we have then

$$\theta = \frac{\lambda(1-z)}{z},$$

and the values of z are given by the equation

$$\lambda(1-z)^2 - \lambda z(1-z) - 2\nu z^2 = 0,$$

that is

$$2(\lambda - \nu)z^2 - 3\lambda z + \lambda = 0,$$

which, putting $\frac{\nu - \lambda}{2\nu} = \varpi$, or $\varpi = 1 - \frac{\lambda}{2\nu}$ (ϖ is the distance of the satellite line from the asymptote $z = 0$), becomes

$$2z^2 - 3\varpi z + \varpi = 0,$$

or we have

$$z = \frac{1}{4}(3\varpi \pm \sqrt{\varpi(9\varpi - 8)}).$$

The condition for a twofold centre is ($\varpi = 0$ which may be disregarded, or else)

$$9\varpi - 8 = 0;$$

or, what is the same thing, $\lambda + 8\nu = 0$.

100. If z_1, z_2 are the coordinates of the two critic centres, then we have

$$2z_1^2 = \varpi(3z_1 - 1), \quad 2z_2^2 = \varpi(3z_2 - 1),$$

and thence

$$\frac{z_1^2}{z_2^2} = \frac{3z_1 - 1}{3z_2 - 1},$$

or, reducing,

$$3z_1z_2 - z_1 - z_2 = 0,$$

or in terms of the x -coordinates

$$3x_1x_2 - 2(x_1 + x_2) + 1 = 0,$$

which equation however merely expresses that the two centres are harmonics to each other in regard to the twofold centre conic $3x^2 - 4x - y^2 + 1 = 0$. It is right to remark that the formulae, although referring to a different system of coordinates, are absolutely identical with those given Nos. 67–69, writing therein ν for μ , and z for x .

101. An inspection of the form of the envelope shows what are the positions of the satellite line which gives rise to Plücker's Groups for the Hyperbolas *A Defective*. We have in fact.

Hyperbolas A Defective, asymptote not osculating.

The satellite line is not parallel to the asymptote, and the different positions give the following six of Plücker's Groups, viz.

XVIII. Satellite line cuts upper, cuts lower, branch of envelope.

XX. „ „ „ „ „ and passes through asymptote point.

XIX. Satellite line touches upper, cuts lower, branch.

XXI. „ „ does not cut upper, cuts lower, branch.

XXIII. „ „ „ „ touches lower branch.

XXII. „ „ „ „ does not cut lower branch.

Hyperbolas A Defective, asymptote osculating.

The satellite line is parallel to the asymptote, and we have the six groups,

XXVIII. Satellite line cuts upper branch, cuts lower branch of envelope, viz. it lies above the asymptote point.

XXIX. Do., Do., but it passes through the asymptote point.

XXX. Do., Do., but it lies below the asymptote point.

XXXI. Satellite line touches upper branch, cuts lower branch.

XXXII. „ „ does not cut upper branch, cuts lower branch.

XXXIII. „ „ „ „ does not cut lower branch, viz. it lies below the asymptote.

And finally,

Hyperbolas A Defective, three osculating asymptotes.

Satellite line at infinity, giving the single group

XXXV.

But the division gives rise to a remark such as is made *ante* No. 71.

As to the Groups of the Hyperbolas B. Article Nos. 102 to 104.

102. Taking $z = 0$ as the equation of the line infinity, and $x = 0, y = 0$ as the equation of any two lines through the point of intersection of the asymptotes, or “asymptote point,” then the equation of the cubic may be taken to be

$$\frac{1}{3}(a, b, c, d \mid x, y)^3 + kz^2(\lambda x + \mu y + \nu z) = 0.$$

103. To determine the critic centres we have

$$\begin{aligned} (a, b, c \mid x, y)^2 + kz^2\lambda &= 0, \\ (b, c, d \mid x, y)^2 + kz^2\mu &= 0, \\ 2z(\lambda x + \mu y + \nu z) + 3z^2\nu &= 0, \end{aligned}$$

and thence

$$\mu(a, b, c \mid x, y)^2 - \lambda(b, c, d \mid x, y)^2 = 0,$$

or as it may also be written

$$(\mu a - \lambda b, \mu b - \lambda c, \mu c - \lambda d \mid x, y)^2 = 0,$$

and also

$$2(\lambda x + \mu y) - 3\nu z = 0,$$

which two equations determine the critic centres for a given position of the satellite line; the first of them gives a pair of lines through the asymptote point; the latter is a line parallel to the satellite line: there are thus two critic centres.

104. The condition for a twofold centre is

$$(ac - b^2, bc - ad, bd - c^2 \mid \lambda, \mu)^2 = 0,$$

so that there are a pair of twofold centres which will be real if

$$(bc - ad)^2 - 4(ac - b^2)(bd - c^2) = +,$$

imaginary if

$$(bc - ad)^2 - 4(ac - b^2)(bd - c^2) = -,$$

that is, the twofold centres will be real or imaginary, according as the equation

$$(a, b, c, d \mid x, y)^3 = 0$$

has its roots one real and two imaginary, or all three real; viz. the twofold centres are real for the Hyperbolas *B* Defective; imaginary for the Hyperbolas *B* Redundant.

And we see also that for the Hyperbolas *B* Redundant the critic centres are always real; but that for the Hyperbolas *B* Defective, they may be both real, or both imaginary, or may coincide together, giving a twofold centre. But the two cases are best studied by assuming different special forms for the equation.

The Hyperbolas B Redundant. Article Nos. 105 to 107.

105. The equation may be taken to be

$$xy(x - y) + kz^2(\lambda x + \mu y + \nu z) = 0,$$

or writing $z = 1$, then the equation is

$$xy(x - y) + k(\lambda x + \mu y + \nu z) = 0.$$

We may, to fix the ideas, consider the case where the three asymptotes are parallel to the sides of an equilateral triangle; x, y , and $x - y$ will then denote the perpendicular distances of the point from the three asymptotes respectively.

106. The critic centres are given by the equations

$$\begin{aligned} 2(\lambda x + \mu y) - 3\nu &= 0, \\ \lambda x^2 - 2(\lambda + \mu)xy + \mu y^2 &= 0; \end{aligned}$$

or, what is the same thing, they are the intersections of the line

$$2(\lambda x + \mu y) - 3\nu = 0$$

by the two real lines

$$\lambda x = [(\lambda + \mu) \pm \sqrt{\lambda^2 + \lambda\mu + \mu^2}] y.$$

107. The groups are

Hyperbolas B Redundant. No osculating asymptote.

The satellite line not parallel to any asymptote, that is $\lambda = 0$, $\mu = 0$, $\lambda + \mu = 0$. We have the two groups

VII. Satellite line does not pass through asymptote point (ν not $= 0$).

VIII. Satellite line passes through asymptote point ($\nu = 0$).

Hyperbolas B Redundant. One osculating asymptote. Satellite line is parallel to an asymptote, suppose to the asymptote $x = 0$; that is, $\mu = 0$, or the satellite line is $\lambda x + \nu = 0$. We have only the group

XV.

Hyperbolas B Redundant. Three osculating asymptotes. Satellite line lies at infinity, that is, $\lambda = 0$, $\mu = 0$. We have only the group

XVII.

The Hyperbolas B Defective. Article Nos. 108 to 110.

108. The equation may be taken to be

$$\frac{1}{3}x(x^2 + y^2) + kz^2(\lambda x + \mu y + \nu z) = 0,$$

or writing $z = 1$, then it is

$$\frac{1}{3}x(x^2 + y^2) + k(\lambda x + \mu y + \nu) = 0,$$

and if to fix the ideas we take the case where the two imaginary asymptotes are the asymptotes of a circle, then x , y will be ordinary rectangular coordinates.

109. The critic centres are given by the equations

$$\begin{aligned} 2(\lambda x + \mu y) - 3\nu &= 0, \\ 3\mu x^2 - 2\lambda xy + \mu y^2 &= 0, \end{aligned}$$

that is they are the intersections of the line $2(\lambda x + \mu y) - 3\nu = 0$ (which is a line parallel to the satellite line, on the other side of the asymptote point and at a distance from it $= \frac{1}{2}$ distance of satellite line) by the pair of lines

$$3\mu x = (\lambda \pm \sqrt{\lambda^2 - 3\mu^2}) y.$$

Hence the critic centres are real if $\lambda^2 > 3\mu^2$, that is, if the satellite line is inclined to the asymptote at an angle $> 60^\circ$; imaginary if $\lambda^2 < 3\mu^2$, that is, if the satellite line is inclined to the asymptote at an angle $< 60^\circ$; and there is a twofold centre if $\lambda^2 = 3\mu^2$, that is, if the inclination is $= 60^\circ$. This assumes, however, that ν is not $= 0$, that is that the satellite line does not pass through the asymptote point; when it does the distinction of the cases disappears. Hence the groups are

110. *Hyperbolas B Defective. No osculating asymptote.* The Satellite line is not parallel to the asymptote, and the groups are,

Satellite line not passing through asymptote point.

XXIV. Satellite line inclined to asymptote at angle $> 60^\circ$.

XXV. „ „ „ „ $= 60^\circ$.

XXVI. „ „ „ „ $< 60^\circ$.

Satellite line passes through asymptote point, the single group

XXVII.

Hyperbolas B Defective. Real osculating asymptote. The satellite line is parallel to the asymptote, and we have the single group

XXXIV.

Hyperbolas B Defective. Three osculating asymptotes. Satellite line is at infinity and we have the single group

XXXVI.

The foregoing theory of the hyperbolas *A* and *B* completes the enumeration of the groups I. to XXXVI.

As to the groups of the parabolic hyperbolas. Article Nos. 111 to 115.

111. I consider the equation in the form

$$\frac{1}{3}x(by^2 + cz^2 + 2gzx) + kz^2(\mu y + \nu z) = 0,$$

viz. the cubic $x(by^2 + cz^2 + 2gzx) = 0$ is made up of a conic $by^2 + cz^2 + 2gzx = 0$, and a line $x = 0$; the other cubic $z^2(\mu y + \nu z) = 0$ is made up of a tangent of the conic, regarded as a twofold line, $z^2 = 0$, and of a line $\mu y + \nu z = 0$ through the point of contact of such tangent.

112. To determine the critic centres we have

$$\begin{aligned} x \cdot gz + \frac{1}{3}(by^2 + cz^2 + 2gzx) &= 0, \\ x \cdot by + kz \cdot \mu z &= 0, \\ x(cz + gx) + kz(2\mu y + 3\nu z) &= 0; \end{aligned}$$

eliminating k from the second and third equations

$$\mu z(cz + gx) - by(2\mu y + 3\nu z) = 0,$$

that is

$$-2b\mu y^2 + c\mu z^2 + g\mu zx - 3b\nu yz = 0;$$

or reducing by means of the first equation written under the form

$$by^2 + cz^2 + 4gzx = 0,$$

we find

$$3c\mu z^2 + 9g\mu zx - 3b\nu yz = 0,$$

that is $z = 0$, which may be rejected, or else

$$c\mu z + 3g\mu x - b\nu y = 0,$$

or, as it may be written,

$$\mu(3gx + cz) - \nu by = 0.$$

Hence the entire series of critic centres lie on the conic

$$by^2 + cz^2 + 4gzx = 0,$$

and corresponding to the satellite line $\mu y + \nu z = 0$, we have the two critic centres which are the intersections of the conic by the line

$$\mu(3gx + cz) - \nu by = 0,$$

the lines pass through the fixed point $3gx + cz = 0$, $y = 0$, and form a pencil homographic with the satellite lines $\mu y + \nu z = 0$.

113. We have a twofold centre if the line touches the conic, the condition for this is

$$(bc, -4g^2, 0, 0, -2bg, 0 \mid 3g\mu, -b\nu, c\mu)^2 = 0,$$

that is $3bg^2(3c\mu^2 + 4b\nu^2) = 0$, or simply,

$$3c\mu^2 + 4b\nu^2 = 0,$$

and from this and the equation $\mu y + \nu z = 0$, eliminating μ and ν we find

$$4by^2 + 3cz^2 = 0,$$

for the equation of the satellite lines which respectively give rise to a twofold centre; the lines in question are real or imaginary according as the lines $by^2 + cz^2 = 0$ are real or imaginary, that is, according as the line $x = 0$ cuts the conic $by^2 + cz^2 + 2gzx = 0$ in two real, or in two imaginary, points.

114. Writing now $b = 1$, $c = -mn$, $2g = n$ and $x + mz$ in the place of z , the equation is

$$(x + mz)(y^2 + nzx) + 2kz^2(\mu y + \nu z) = 0,$$

and we may consider $z = 0$ as the equation of the line infinity: writing in the formulae $z = 1$, the critic centres are given as the intersection of the parabola

$$y^2 + 2nx + mn = 0,$$

with the line

$$\nu y = \frac{1}{4}\mu n(3x + m);$$

and the condition for a two-fold centre is

$$4\nu^2 - 3mn\mu^2 = 0;$$

the equation of the satellite lines corresponding respectively to a two-fold centre is

$$4y^2 - 3mn = 0;$$

the lines are real or imaginary according as mn is positive or negative, or (observing that the equations $x + m = 0$, $y^2 + nx = 0$ give $y^2 - mn = 0$), according as the line $x + m = 0$ cuts or does not cut the parabola $y^2 + nx = 0$. Suppose for a moment that the line does cut the parabola and that y_1 is the corresponding value of y , then $y_1^2 = mn$; and the equation $4y^2 - 3mn = 0$ of the satellite lines which correspond respectively to the case of a two-fold centre is $y^2 = \frac{3}{4}y_1^2$. We have thus $y = \pm y_1$ and $y = \pm\sqrt{\frac{3}{4}}y_1$ as special positions of the satellite line $\mu y + \nu = 0$. In the case where the line $x + m = 0$ touches the parabola $y^2 + nx = 0$, the value of y_1 is $= 0$, and we have only the special position $y = 0$; finally, when the line does not cut the parabola there is no special position.

115. Plücker's groups are consequently as follows:

Parabolic Hyperbolas; ordinary asymptote and five-pointic asymptotic parabola, that is the line $\mu y + \nu = 0$ is not at infinity.

Asymptote cuts parabola, $mn = +$.

XXXVII. Satellite line lies outside the lines $y = \pm y_1$ which belong to the points of intersection.

XXXVIII. Satellite line passes through a point of intersection, that is, coincides with one of the lines $y = \pm y_1$.

XXXIX. Satellite line lies between the lines $y = \pm y_1$ and $y = \pm\sqrt{\frac{3}{4}}y_1$.

XL. Satellite line coincides with one of the lines $y = \pm\sqrt{\frac{3}{4}}y_1$, which give respectively a two-fold centre.

XLI. Satellite line lies between the lines $y = \pm\sqrt{\frac{3}{4}}y_1$.

Asymptote touches parabola, viz. $m = 0$.

XLIII. Satellite line does not pass through the point of contact.

XLIV. Satellite line passes through point of contact or its equation is $y = 0$.

Asymptote does not cut parabola, viz. $mn = -$. This gives the single group

XLII.

Parabolic hyperbolas. Osculating asymptote and six-pointic asymptotic parabola. The satellite line is here at infinity, and there is no new distinction of groups. The groups therefore are

Asymptote cuts parabola,

XLV.

Asymptote does not cut parabola,

XLVI.

Asymptote touches parabola,

XLVII.

As to the Groups of the Central and Parabolic Hyperbolisms. Article No. 116.

116. For the Hyperbolisms, Central and Parabolic, since these have a node or a cusp at infinity, they cannot acquire a new node, and the theory of critic centres does not arise. There is, however, as regards the Hyperbolisms of the Hyperbola a distinction in the position of the satellite line, viz. this may lie outside, or between, the parallel asymptotes. The groups are

Hyperbolisms of the Hyperbola. Ordinary asymptote. The satellite line is not at infinity, and it may lie in either of the positions just mentioned. We have therefore

XLVIII. Satellite line lies between the parallel asymptotes.

XLIX. „ „ outside „ „ .

Osculating asymptote; the satellite line is at infinity. We have

L.

Hyperbolisms of the Ellipse. Ordinary asymptote. The satellite line is not at infinity, and we have

LI.

Osculating asymptote. Satellite line is at infinity,

LII.

Hyperbolisms of the Parabola. Ordinary asymptote. Satellite line is not at infinity, we have

LIII.

Osculating asymptote; satellite line is at infinity; we have

LIV.

As to the Groups of the Divergent Parabolas. Article Nos. 117 and 118.

117. Taking the equation under the form

$$ax^3 + by^2z + kz^2(\lambda x + \nu z) = 0,$$

we find for a critic centre

$$\begin{aligned} 3ax^2 + kz \cdot \lambda z &= 0, \\ 2byz &= 0, \\ by^2 + kz(2\lambda x + 3\nu z) &= 0; \end{aligned}$$

hence there is a single critic centre $y = 0$, $2\lambda x + 3\nu z = 0$, the critic value of k is $k = -\frac{27a\nu^2}{4\lambda^3}$, and with this value of k the equation in fact is

$$4\lambda^3(ax^3 + by^2z) - 27a\nu^2(\lambda x + \nu z)z^2 = 0,$$

that is

$$a(4\lambda^3x^3 - 27\lambda\nu^2xz^2 - 27\nu^3z^3) + 4\lambda^3by^2z = 0,$$

or, as it may be written,

$$a(2\lambda x + 3\nu z)^2(\lambda x - 3\nu z) + 4\lambda^3by^2z = 0,$$

which puts in evidence the critic centre or node of the curve. But, as there is here only a single critic centre, there is of course no further theory of the two-fold centre, &c.

118. The groups are as given a priori by the relation of the satellite line $\lambda x + \nu z = 0$, to the semicubical parabola $ax^3 + by^2z = 0$, viz. writing $z = 1$ and changing the constants,

Divergent Parabolas, the semicubical parabola $y^2 = x^3$, which is

LV.

Divergent Parabolas, Asymptotic Semicubical Parabola of seven-pointic contact, viz. equation of the asymptotic parabola being $y^2 - x^3 = 0$, and writing for convenience $k(\lambda x + \nu) = -3ax + 2b = 0$ for the equation of the satellite line, the equation of the curve is $y^2 = x^3 - 3ax + 2b$. And the groups are

LVI. Satellite line cuts asymptotic parabola.

LVII. „ does not cut „ „ .

LVIII. „ passes through vertex of parabola.

And further

Divergent Parabolas. Asymptotic Semicubical Parabola of nine-pointic intersection. The satellite line is at infinity, and the equation is $y^2 = x^3 + 2b$. This is group

LIX.

As to the Trident Curve and the Cubical Parabola. Article Nos. 119 and 120.

119. For the Trident Curve, equation is $x(x^2 + \lambda y) + \mu = 0$, or the satellite line is at infinity, and there is no distinction of groups; we have only group

LX.

120. For the Cubical Parabola this is $x^3 + \mu y = 0$, there is no distinction of groups, and the curve is group

LXI.

As to the Division into Species: Comparison of Newton and Plücker. Article Nos. 121 and 122.

121. The division into species is obtained without difficulty when the groups are once established; in fact it only remains to trace for each given form of V and s the series of curves $V + \mu s = 0$, as μ passes from ∞ to $-\infty$ through the value 0 and the critic values which correspond to nodal curves: I have nothing to add to what has been done by Plücker, and it is unnecessary to reproduce the investigation. It may be remarked

that the mere inspection of Plücker's figures is sufficient to show which of his species correspond to the same Newtonian species; the species which do so belong to the same Newtonian species in some instances closely resemble each other in form, although in others the difference of form is apparent enough: but the Plückerian species which correspond to the same Newtonian species belong for the most part to different groups and are thus distinguished from each other by the characters which distinguish the groups to which they respectively belong: thus for instance Newton's Species 1 (a hyperbola *A Redundant*) is characterised as consisting of three hyperbolic branches, one inscribed, one circumscribed, and one ambigène, with an oval. Such a curve may exist with three different positions of the satellite line in regard to the asymptotes, viz. the satellite line may cut the three sides produced, or it may pass through a vertex, cutting the opposite side produced, or it may cut two sides and the third side produced, not cutting the envelope — which are the characters of Plücker's groups I, II, IV, respectively, and there belongs to the Newtonian species 1, a species out of each of these groups, viz. they are I. 1; II. 9, and IV. 18.

122. The correspondence of the Plückerian Species with those of Newton is shown in the following Table.

Newton's Genus 1, contains 9 Species, viz.

	1	2	3	4	5	6	7	8	9
corresponding to Plücker's Species									
I.	1			3					
II.		2		8	7	4		5, 6	
III.			9	10			12		11
IV.	17			19	16, 20			22, 23	
V.					24, 26	25	31	28, 29	
VI.	18							32, 33	

Part of Newton's Genus 4, contains 3 Species, viz.

	24	25	26
corresponding to Plücker's Species			
VII.	34	35	36
VIII.		37	

Newton's Genus 2, contains 14 Species, viz.

	10	10'	11	12	13	13'	14	15	16	17	18	19	20	21
corresponding to Plücker's Species														
IX.	38		39		40							47		
X.		44	45			46		41	42		43			
XI.	50				48		52				51			
XII.		49	53			54	56				55			
XIII.	60		61					62	65		63			
XIV.														64

Further part of Newton's Genus 4, contains 4 Species, viz.

	28	29	30	31
corresponding to Plücker's Species				
XV.	66	67	68	69

Newton's Genus 3, contains 4 Species, viz.

	22	22'	22''	23
corresponding to Plücker's Species				
XVI.	70	72	71	73

Residue of Newton's Genus 4, contains 1 Species, viz.

	32
corresponding to Plücker's Species	
XVII.	74

Newton’s Genus 5, contains 6 Species, viz.

	33	34	35	36	37	38
corresponding to Plücker’s Species						
XVIII.	77, 82	76, 81	84		75, 79, 80	
XIX.	88	87	90		83, 85, 86	
XX.	89				94, 95	
XXI.	92	93	107	99	98, 102, 103	
XXII.	96, 100	97, 101	117	111	106, 108, 109	
XXIII.	104	105	113		110, 114, 115	
XXIV.	112					
XXV.					116, 118, 119	
XXVI.						
XXVII.						121

Newton’s Genus 6, contains 7 Species, viz.

	39	40	41	42	43	44	45
corresponding to Plücker’s Species							
XXVIII.	122	123, 128	131, 135	122, 129	126	124, 130	127
XXIX.		137	138			132, 136	
XXX.		141		139	133	138, 142	140
XXXI.	143	148	145				
XXXII.		152					
XXXIII.	144					146, 149	
XXXIV.						150, 153	
XXXV.	154		155			156, 158	157
XXXVI.							

Newton’s Genus 7, contains 7 Species, viz.

	46	47	48	49	50	51	52
corresponding to Plücker’s Species							
XXXVII.	159		160	166	161, 162	163	164
XXXVIII.	165			172	167, 168		
XXXIX.	171	170			169, 173, 174		
XL.					179		
XLI.		176			175, 177, 178		
XLII.					182	181	180
XLIII.					185, 187	186	183
XLIV.						184	

Newton’s Genus 8, contains 6 Species, viz.

	53	54	55	56	56’	56’’
corresponding to Plücker’s Species						
XLV.	190	193	191	188	189	
XLVI.			197	195		
XLVII.	196		192			

Newton’s Genus 9, contains 4 Species, viz.

	57	58	59	60
corresponding to Plücker’s Species				
XLVIII.	199	198		
XLIX.				
L.			200	

Newton’s Genus 10, contains 3 Species, viz.

	61	62	63
corresponding to Plücker’s Species			
LI.	201		
LII.		202	

Newton’s Genus 11, contains 2 Species, viz.

	64	65
corresponding to Plücker’s Species		
LIII.	203	
LIV.		204

Newton’s Genus 12, contains 1 Species, viz.

	66
corresponding to Plücker’s Species	
LX.	218

Newton’s Genus 13, contains 5 Species, viz.

	67	68	69	70	71
corresponding to Plücker’s Species					
LV.	208	207		205	206, 209
LVI.	212		211		210, 213
LVII.	214				215
LVIII.					216, 217
LIX.					

Newton’s Genus 14, contains 1 Species, viz.

	72
corresponding to Plücker’s Species	
LXI.	219

It is to be noticed that (as appears by the Table) Plücker enumerates 13 species of the Divergent Parabola, viz. corresponding to the Parabola Pura of Newton we have five species, and to the Parabola cum Ovali three species; but to each of the other three Newtonian species (Nodata, Punctata, Cuspidata) only a single species. The difference in nowise affects Newton’s before-mentioned theorem, that every cubic curve is the shadow of a Divergent Parabola; but (the characters of Plücker’s species being unaffected by projection) the number of resulting kinds of cubic curves (or cones) will be five or thirteen according as the one or the other classification is adopted; but this is a subject which I do not enter upon in the present Memoir.

Cambridge, February 8, 1864.

[Figure: Plate III, illustrating the Hyperbolas A Defective discussed in Arts. 73–100 above: the envelope (an upper branch crossing itself at a crunode together with a separate lower branch), the twofold-centre hyperbola, the one-with-twofold-centre locus, and a representative harmonic conic (here a flat ellipse), with marked points A, C, D, E, M, N and their critic-centre images on the harmonic conic.]